Symmetrical Components & Sequence Networks

From unbalanced faults to solvable circuits — a step-by-step visual guide to one of power engineering's most powerful analytical tools.

The Core Trick: Three "Balanced Worlds" Inside One Unbalanced System

Any unbalanced 3-phase set can be mathematically decomposed into three perfectly balanced sequences. Think of it like a prism splitting white light into three pure colours — each sequence travels independently and recombines to recreate the original messy signal.

Positive Sequence

Balanced, ABC rotation. The "normal" operating world. Like three friends walking in step.

Negative Sequence

Balanced, ACB rotation — spinning backwards. Indicates asymmetric disturbance or fault.

Zero Sequence

All three phases in phase with each other. The "earth-return" world — only exists when current flows to ground.

Key principle: **Va = V1 + V2 + V0** — every phase quantity is the superposition of three sequence components.

Introduction to Symmetrical Components

An **unbalanced 3-phase system** can be represented as the sum of **three balanced sets of components**:

$$\boxed{\text{Unbalanced 3-}\phi = \text{Positive Sequence} + \text{Negative Sequence} + \text{Zero Sequence}}$$

Positive Sequence (1)

- Three equal phasors, 120° apart
- Normal phase order: **a-b-c**
- Represents the **normal balanced component**

Negative Sequence (2)

- Three equal phasors, 120° apart
- Reverse phase order: **a-c-b**
- Appears due to **system unbalance**

Zero Sequence & Application to Fault Analysis

Zero Sequence (0)

- Three equal phasors
- All three phasors are **in phase**

$$\boxed{V_{a0} = V_{b0} = V_{c0}}$$

- Particularly important in **faults involving ground**

Combining the Sequence Components

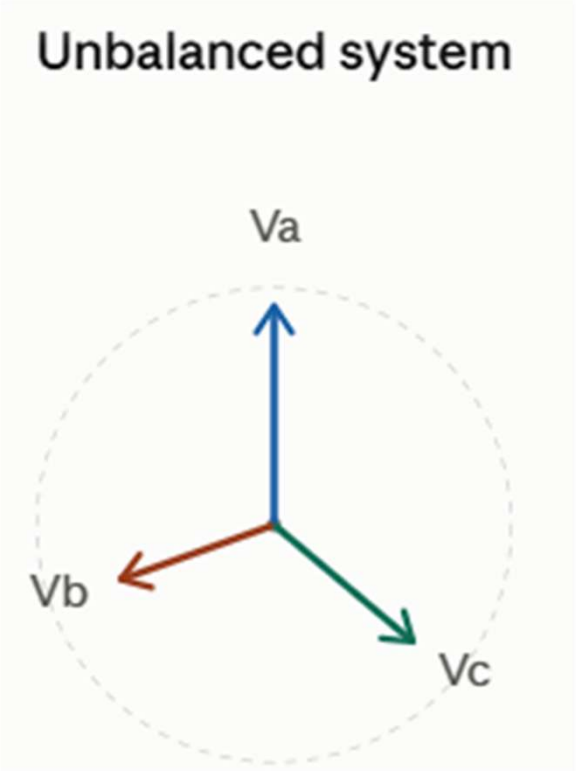
$$\boxed{V_a = V_{a1} + V_{a2} + V_{a0}}$$

Similarly for phases b and c.

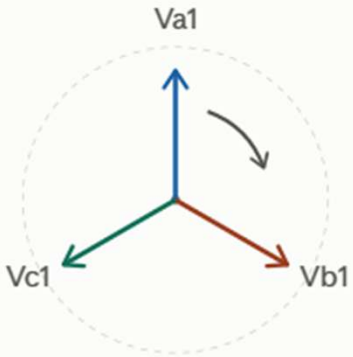
Key Idea

$$\boxed{\text{Unbalanced Fault} \rightarrow \text{Sequence Networks} \rightarrow \text{Simpler Analysis}}$$

Symmetrical components allow unbalanced faults to be analyzed using balanced sequence networks.

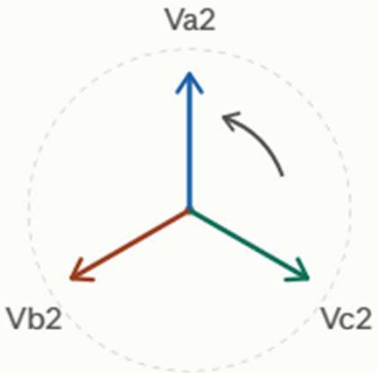

Unbalanced system
Va
Vb
Vc

Positive sequence



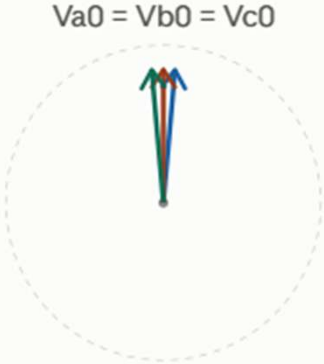
Rotates a-b-c, same order as the original system

Negative sequence



Rotates a-c-b, reversed order

Zero sequence



Equal magnitude, in phase, no rotation

Sequence Components Under Balanced Conditions

Balanced 3-Phase System

For a perfectly balanced three-phase system:

$$V_a = V\angle 0^\circ, \qquad V_b = V\angle -120^\circ, \qquad V_c = V\angle 120^\circ$$

Sequence Components

Positive Sequence (V_1) — EXISTS

- Equal magnitudes
- 120° apart
- Normal phase sequence: **a-b-c**

Negative Sequence (V_2) — ZERO

- Appears when the system is unbalanced

Zero Sequence (V_0) — ZERO

- Appears under certain unbalanced conditions, particularly ground faults

$$\boxed{V_1 = V, \qquad V_2 = 0, \qquad V_0 = 0}$$

Sequence Components in Fault Analysis

Balanced 3-Phase Fault

A three-phase fault is a **balanced fault**.

Therefore,

$$\boxed{\text{Balanced 3-Phase Fault} \Rightarrow \text{Positive-Sequence Network Only}}$$

Hence,

$$\boxed{V_2 = 0, \qquad V_0 = 0}$$

Unbalanced Faults

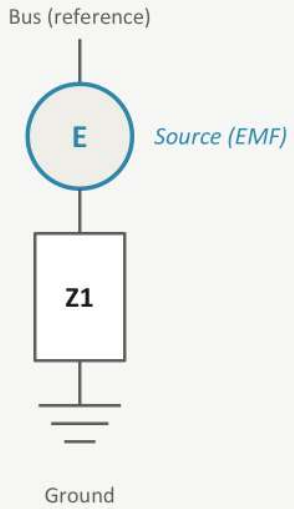
Negative- and/or zero-sequence components appear during **unbalanced faults**.

- **Line-to-Line (L-L) fault:** Positive + Negative sequence
- **Single Line-to-Ground (L-G) fault:** Positive + Negative + Zero sequence
- **Double Line-to-Ground (L-L-G) fault:** Positive + Negative + Zero sequence

Key Point

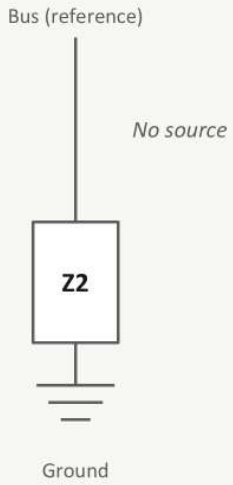
$$\boxed{\text{Negative and Zero Sequences} \Rightarrow \text{Used for Unbalanced Fault Analysis}}$$

Positive-sequence network



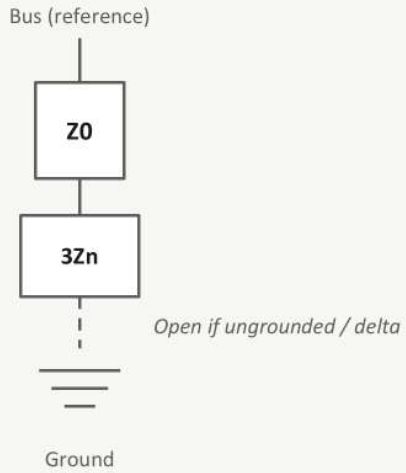
- Looks like the normal single-line diagram used for load flow
- Includes EMF sources — generator internal voltages drive this network
- Impedances: generator reactance, transformer leakage reactance, line impedance
- Neutral / ground is the reference bus
- The only network with sources under balanced operation

Negative-sequence network



- Same static impedances as positive sequence — lines and transformers don't care about phase rotation
- No EMF sources — machines don't generate negative-sequence voltage internally
- Rotating machines use X2, usually close to the subtransient reactance
- Reference bus is grounded but carries no source voltage

0 Zero-sequence network



- Topology depends on grounding and transformer connections
- Needs a physical path through neutral to flow
- Grounded-Y: path exists — model 3×Zn in series with neutral
- Ungrounded-Y or delta: path is open — current can't escape that side
- No EMF sources

Actual Phase Quantities → Sequence Components

Given the actual phase quantities $V_a,\ V_b,\ V_c$:

$$\boxed{\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}}$$

Therefore,

$$V_0 = \frac{1}{3}(V_a + V_b + V_c)$$

$$V_1 = \frac{1}{3}(V_a + aV_b + a^2V_c)$$

$$V_2 = \frac{1}{3}(V_a + a^2V_b + aV_c)$$

where

$$\boxed{a = 1\angle 120^\circ, \qquad a^2 = 1\angle 240^\circ, \qquad 1 + a + a^2 = 0}$$

Sequence Components → Actual Phase Quantities

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix}$$

Hence,

$$V_a = V_0 + V_1 + V_2$$

$$V_b = V_0 + a^2 V_1 + a V_2$$

$$V_c = V_0 + a V_1 + a^2 V_2$$

The same transformation applies to currents: simply replace V by I

Approach to Fault Problems Using Sequence Networks

1. Convert All Impedances to a Common PU Base

Determine:

$$\boxed{Z_1, \qquad Z_2, \qquad Z_0}$$

2. Identify the Type of Fault

Fault	Sequence Networks Required
3-phase fault	Positive only
L-L fault	Positive + Negative
L-G fault	Positive + Negative + Zero
L-L-G fault	Positive + Negative + Zero

Overall Problem-Solving Flow

One-Line Diagram

↓

Select Common Base

↓

Convert All Data to PU

↓

Construct $(+), (-), (0)$ Sequence Networks

↓

Reduce Networks to Fault Point

↓

Connect Networks According to Fault

↓

Calculate I_0, I_1, I_2

↓

Find Actual Phase Currents / Voltages

The phase voltages are

$$V_a = 100\angle 0^\circ, \qquad V_b = 100\angle -120^\circ, \qquad V_c = 100\angle 120^\circ \text{ V}$$

Find the symmetrical components.

apply the transformation matrix:

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 100 \\ 100a^2 \\ 100a \end{bmatrix}$$

Multiplying,

$$= \frac{100}{3} \begin{bmatrix} 1 + a + a^2 \\ 1 + a^3 + a^3 \\ 1 + a^4 + a^2 \end{bmatrix}$$

Using

$$a^3 = 1, \qquad 1 + a + a^2 = 0$$

we get

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 100 \\ 0 \end{bmatrix}$$

Therefore,

$$\boxed{V_0 = 0, \qquad V_1 = 100\angle 0^\circ, \qquad V_2 = 0}$$

Given

$$V_a = 100\angle 0^\circ, \qquad V_b = 100\angle 120^\circ, \qquad V_c = 100\angle -120^\circ \text{ V}$$

Find the symmetrical components.

Write

$$V_b = 100a, \qquad V_c = 100a^2$$

Then

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 100 \\ 100a \\ 100a^2 \end{bmatrix}$$

which gives

$$\boxed{\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 100 \end{bmatrix}}$$

Hence,

$$\boxed{V_0 = 0, \qquad V_1 = 0, \qquad V_2 = 100\angle 0^\circ}$$

Given

$$V_a = V_b = V_c = 60\angle 0^\circ \text{ V}$$

Find the symmetrical components.

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 60 \\ 60 \\ 60 \end{bmatrix}$$

Therefore,

$$= \frac{60}{3} \begin{bmatrix} 3 \\ 1 + a + a^2 \\ 1 + a + a^2 \end{bmatrix}$$

Hence,

$$\boxed{\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix}}$$

Therefore,

$$\boxed{V_0 = 60\angle 0^\circ, \qquad V_1 = 0, \qquad V_2 = 0}$$

$$V_a = 100\angle 0^\circ \text{ V}$$

$$V_b = 80\angle -120^\circ \text{ V}$$

$$V_c = 60\angle 120^\circ \text{ V}$$

Find $V_0,\ V_1,\ V_2$.

Write

$$V_a = 100, \qquad V_b = 80a^2, \qquad V_c = 60a$$

Use

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} 100 \\ 80a^2 \\ 60a \end{bmatrix}$$

This gives

$$V_0 = \frac{1}{3}(100 + 80a^2 + 60a)$$

$$V_1 = \frac{1}{3}(100 + 80 + 60)$$

$$V_2 = \frac{1}{3}(100 + 80a + 60a^2)$$

Therefore,

$$\boxed{V_1 = 80\angle 0^\circ \text{ V}}$$

and evaluating the other two,

$$\boxed{V_0 = 11.55\angle -30^\circ \text{ V}}$$

$$\boxed{V_2 = 11.55\angle 30^\circ \text{ V}}$$

So,

$$\boxed{V_0 = 11.55\angle -30^\circ, \quad V_1 = 80\angle 0^\circ, \quad V_2 = 11.55\angle 30^\circ}$$

Given

$$I_0 = 2\angle 0^\circ, \qquad I_1 = 10\angle 0^\circ, \qquad I_2 = 4\angle 0^\circ \text{ A}$$

find I_a, I_b, I_c.

Use:

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \\ 4 \end{bmatrix}$$

Multiplying,

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 2 + 10 + 4 \\ 2 + 10a^2 + 4a \\ 2 + 10a + 4a^2 \end{bmatrix}$$

Using

$$a = -0.5 + j0.866, \qquad a^2 = -0.5 - j0.866$$

gives

$$\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 16 \\ -5 - j5.196 \\ -5 + j5.196 \end{bmatrix}$$

$$\boxed{I_a = 16\angle 0^\circ \text{ A}}$$

$$\boxed{I_b = 7.21\angle -133.9^\circ \text{ A}}$$

$$\boxed{I_c = 7.21\angle 133.9^\circ \text{ A}}$$

Conclusion: The Grid's Symmetrical Backbone

From slide-rule calculations to millisecond digital protection — symmetrical components remain the universal language of power system fault analysis.

The Core Insight

Any unbalanced fault, no matter how complex, reduces to three independent, solvable single-phase circuits — connected by precise mathematical rules at one point.

The Engineering Value

Sequence networks underpin every protective relay setting, every fault-level calculation, and every stability study on the modern grid.

The Future Challenge

As inverter-based renewables dominate generation, their different fault signatures challenge traditional sequence-based methods — demanding new understanding whilst building on these same foundations.

Master the three sequences and you hold the key to understanding every fault, every relay, and every stability limit on the power network.